RV College of Engineering®, Bengaluru – 59
(Autonomous Institution affiliated to VTU, Belagavi)

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Academic year 2025-2026 (Even Semester)

DEPARTMENT OF
AEROSPACE ENGINEERING

Date	**19th June 2026**	**Maximum Marks**	**60**
Course Code	**AS266TEA**	**Duration**	**120 Min**
Sem	**VI Semester**	**CIE-3**	
FUNDAMENTALS OF AEROSPACE ENGINEERING			

Note: 1. Answer All Questions from both Part-A & Part-B
2. Part-A to be answered in the 1st two pages of the answer booklet

PART – A **(10 Marks)**

	QUESTIONS	M	CO	BL
1.	Which orbital element determines the shape of an orbit?	01	CO2	L2
2.	What is the type of orbit transfer, which is considered the most fuel-efficient orbital transfer method.	01	CO2	L2
3.	Bulkheads are generally oriented in the _____ direction.	01	CO2	L1
4.	Truss structures are most efficient in resisting _____ loads	01	CO2	L2
5.	Which structural member divides the fuselage into compartments?	01	CO2	L1
6.	Name any two structural members used in semi-monocoque construction.	01	CO2	L1
7.	A satellite has a perigee radius of 7000 km and an apogee radius of 9000 km. Find the semi-major axis.	01	CO2	L3
8.	An aircraft's airspeed indicator receives a pitot pressure of 145 kPa and a static pressure of 120 kPa. Determine the dynamic pressure acting on the aircraft.	01	CO4	L3
9.	A satellite moves from perigee to apogee. Does its velocity increase or decrease? Comment on the Gravitational potential Energy and Kinetic energy at Perigee and apogee.	01	CO4	L4
10.	An aircraft climbs from 2500 ft to 8500 ft in 6 minutes. Determine the average rate of climb.	01	CO4	L3

PART-B

Sl. No.	Questions	M	CO	BT
1	State and explain the 3 Kepler's Laws of planetary motion. Mathematically provide the expressions for the 2nd and 3rd Law.	10	CO2	L2

Academic year 2025-2026 (Even Semester)

2	An artificial satellite is launched into a circular orbit at an altitude of 200 km above the Earth's surface. Assume: • Radius of Earth, R_E = 6378km • Mass of Earth, M_E = 5.97 × 10^{24}kg • Universal gravitational constant, G = 6.67 × 10^{-11}N·m^2/kg^2 (a) Calculate the orbital velocity of the satellite. (b) Determine the orbital period of revolution of the satellite. (c) Derive the expression for the radius of a geosynchronous orbit and calculate its value assuming the Earth's rotational period is 86,164 s. (d) Calculate the escape velocity from the Earth's surface and compare it with the orbital velocity obtained in part (a).	10	CO3	L3
3	A spacecraft is initially in a circular Earth orbit at an altitude of 150 miles above the Earth's surface. It is required to transfer to another circular orbit at an altitude of 500 miles using a Hohmann transfer orbit. (a) Using the conservation of angular momentum, prove that $$R_p V_p = R_a V_a$$ where R_p and R_a are the perigee and apogee radii of the transfer orbit, and V_p and V_a are the corresponding velocities. (b) Using conservation of energy, derive expressions for the velocity at perigee and apogee of an elliptical orbit: $$V_p = \sqrt{\frac{2GMR_a}{R_p(R_a+R_p)}} \qquad V_a = \sqrt{\frac{2GMR_p}{R_a(R_a+R_p)}}$$ (c) Determine the velocity increments (ΔV_1 and ΔV_2) required to transfer the spacecraft from the 150-mile orbit to the 500-mile orbit.	10	CO3	L3
4	With a neat diagram of the Pitot-Static System, explain the working principles of the Airspeed Indicator (ASI), Altimeter, and Vertical Speed Indicator (VSI). Discuss the effect of pitot tube blockage and static port blockage on instrument indications	10	CO4	L4
5	With neat sketches, explain the construction and load-carrying mechanisms of Truss, Monocoque, and Semi-Monocoque aircraft structures. Compare them with respect to strength, weight, damage tolerance, and suitability for modern aircraft applications.	10	CO2	L3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	0	27	20	13	3	13	33	11	–	–

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Academic year 2025-2026 (Even Sem)

Common to all branches

DEPARTMENT OF
BIOTECHNOLOGY

Date	**June, 2026**	**Maximum Marks**	**10 + 50**
Course Code	**BT266TEB**	**Duration**	**120 min**
Sem	**VI Semester**	**Closed Book Offline Quiz and Test – 3**	
Healthcare Analytics (Institutional Elective)			

Answer all questions in Part A & Part B, Answer Part A in first two pages in sequence only

Part A

S No	Questions	M	BT	CO
1	What are sequencing adapters in Next-Generation Sequencing (NGS)?	2	2	1
2	What does a Phred quality score of 30 signify in DNA sequencing data?	2	2	1
3	Name any two commonly employed DNA enrichment methods used in NGS workflows.	2	2	2
4	What is meant by the term “base calling” in the context of Next-Generation Sequencing?	2	3	2
5	What is one significant limitation associated with read clipping during NGS data preprocessing?	2	2	3

Part B

S No	Questions (Test)	M	BT	CO
1.	Critically examine the role of quality assessment in NGS data analysis. How are Phred quality scores calculated, and what is their importance in ensuring reliable downstream bioinformatics analyses?	10	2	3
2	Discuss on the evolution of DNA sequencing technologies with special reference to the development of Sanger sequencing.	10	3	3
3	Elucidate on the sources and consequences of adapter and primer contamination in NGS datasets.	10	4	4
4	Compare first-generation and next-generation sequencing technologies in terms of methodology, scalability, accuracy, throughput, and applications in genomic research.	10	4	4
5	Discuss the implications for genomics research and personalized medicine.	10	2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks			60	40		40	20	40		

1. *(Include Nobel Prize, chain termination principle, Human Genome Project impact, etc.)*

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Department of Chemical Engineering

Date: 19.06.2026	**Improvement Test**	**Max. Marks: 60**
Semester: 6	**UG**	**Duration: 2 Hrs**
Course Title: Industrial Safety Engineering		**Course Code: CH266TEC**

S No	Questions	M	BT	CO
	PART A			
1	Mention two employee responsibilities in process industry.	2	1	2
2	Write the types of body protection PPE kits and mention their use.	2	1	4
3	How many MIC storage tanks were installed at the Union Carbide Bhopal plant, and how many were designated for normal operation and emergency use?	2	1	4
4	What were direct casualties reported in Chernobyl disaster?	2	2	4
5	What is the third-year present value factor, up to three decimal places, if a 7% risk-free rate of interest is used to calculate the Net Present Value?	2	3	3
	PART B			
1	Describe the potential incidents of eye and face related hazards. Classify and explain the eye and face PPE kits with sketches.	10	3	4
2	Mention the need of protective equipment for industrial works. Explain six-foot protection personal protective equipment's with sketches.	10	2	4
3	Explain briefly the process chemistry of union carbide Bhopal plant along with the process flow sheet.	10	3	4
4	The table below presents the possible net cash flows and their associated probabilities for Projects X and Y. Assuming a discount rate of 10% and an initial investment of Rs. 5,000 for both projects, calculate the expected Net Present Value (NPV) for each project. Additionally, compute the variance, standard deviation, and coefficient of variation. Based on your analysis, indicate which project is preferable.	20	4	3

Possible events	Project X		Project Y	
	CF	P	CF	P
A	4,000	0.1	12,000	0.10
B	5,000	0.2	10,000	0.15
C	6,000	0.4	8,000	0.50
D	7,000	0.2	6,000	0.15
E	8,000	0.1	4,000	0.10

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Department of Civil Engineering

Date: 16-06-2026	Improvement CIE	Max. Marks: 60
Semester: VI	UG	Duration: 2 Hrs
Course Title: Intelligent Transport Systems		Course Code: CV266TEE

Quiz

S No	Questions	M	BT	CO
1.	Explain the purpose of ITS evaluation in transportation projects.	2	2	3
2.	List the benefits provided by different ITS components in traffic management.	2	2	4
3.	Explain how ITS supports law enforcement in enforcing traffic rules and regulations.	2	2	3
4.	List the various funding options available for ITS implementation.	2	2	4
5.	List the importance of ITS standards testing before deployment.	2	2	3

Test

S No	Questions	M	BT	CO
1.	Explain the various stages involved in ITS evaluation, including project selection, deployment tracking, impact assessment, and evaluation guidelines.	10	3	3
2.	Discuss the benefits of ITS components and explain how they contribute to improving transportation system performance.	10	3	3
3.	Explain the role of ITS in enhancing and supporting the enforcement of traffic rules and regulations with suitable examples.	10	3	4
4.	Explain the application areas of ITS standards and discuss the role of National Transportation Communications for ITS Protocol (NTCIP) in ITS systems.	10	3	4
5.	Discuss the role of ITS in the development of smart cities and explain how ITS applications improve urban mobility, safety, and sustainability.	10	3	4

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Department of Civil Engineering

Date: 19.06.2026	**Improvement CIE**	**Max. Marks: Q10 + T50**
Semester: VI	**UG**	**Duration: 2 Hrs**
Course Title: Integrated Health Monitoring of Structures (Institutional Elective)		**Course Code: CV266TEF**

Note:

i) Answer all the questions.

ii) Use illustrative sketches to substantiate answers when necessary.

PART A

Sl. No.	Questions	M	BT	CO
1.	List any two sensors used in Remote Structural Health Monitoring.	02	L1	CO4
2.	List any two limitations of Remote Structural Health Monitoring.	02	L1	CO4
3.	Discuss the role of IoT in Remote Structural Health Monitoring?	02	L2	CO4
4.	List the advantages of dynamic response methods.	02	L1	CO3
5.	Discuss the role of data acquisition systems in dam monitoring?	02	L2	CO3

PART B

Sl. No.	Questions	M	BT	CO
1.	List and discuss the types of Dynamic field tests.	10	L2	CO3
2.	Discuss the advantages of Remote structural health monitoring.	10	L2	CO4
3.	Discuss the applications of Dynamic Field Testing.	10	L2	CO3
4.	Describe the various methods of Non-Destructive Evaluation used in structural health monitoring and condition assessment.	10	L2	CO4
5.	Explain the various dynamic response methods used for structural condition assessment.	10	L2	CO3

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	–	–	34	26	06	54	–	–	–	–

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Department of Chemistry

Date: 19-06-2026	**Improvement CIE**	**Max. Marks: 60 (50+10)**
Semester: 6	**UG**	**Duration: 2Hrs (30+90 min)**
Course Title: ADVANCED ENERGY STORAGE FOR E-MOBILITY		**Course Code: CM266TEG**

Instructions to candidate

i) All questions are compulsory ii) Part A questions should be answered in first two pages only

PART A

Sl NO	Questions	M	BT	CO
1	State any one limitation of lithium-ion batteries in electric vehicles.	1	2	1
2	What is the electrolyte used in a lead-acid battery?	1	2	1
3	Zebra battery generally operated at high temperature. Justify.	1	3	3
4	Name any one anode materials used in sodium ion batteries.	1	2	1
5	Write the cathode reaction in sodium air batteries when aqueous electrolytes are used.	1	2	4
6	Give an example for asymmetric capacitor.	1	2	1
7	Justify the role of membrane separator in redox flow batteries.	1	3	4
8	What is regenerative braking?	1	2	2
9	Write any one limitation of solar cell hybrid vehicles.	1	3	2
10	Identify any one advantages of fuel cell hybrid vehicles.	1	2	2

PART B

S No	Questions	M	BT	CO
1	With neat labelled diagram outline the construction and working of Nickel metal hydride battery with necessary reactions. List any three advantages and limitations.	10	2	3
2	Illustrate the construction and working of following batteries with respect to e mobility. i) Sodium–Sulphur battery ii) Types of sodium ion batteries with neat labelled diagram.	10	2	1
3	Explain the construction and working of Zebra battery along with its advantages and limitations. Comment on its efficiency over other batteries.	10	2	4
4	i) Outline the construction and working of pseudo type super capacitor. Why pseudo type supercapacitors are superior than that of EDLC? ii) Detail the working mechanism of battery supercapacitor hybrid electric vehicle along with flow chart.	10	4	2
5	Alkali electrolytes are not preferred in methanol oxygen fuel cell. Justify. Outline the construction and working of amorphous silicon solar cell and methanol oxygen fuel cell with neat labelled diagram.	10	2	2

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DEPARTMENT OF MECHANICAL ENGINEERING

Date: 17-06-2026	Improvement CIE	Max. Marks: 60
Semester: VI	UG	Duration: 2 Hrs
Course Title: Automotive Mechatronics (Institutional Elective)		Course Code: ME266TES

QUIZ

#	Question	M	BT	CO
1	Telematics is a combination of telecommunications and _____.	1	1	1
2	Radio communication in vehicles uses _____ waves.	1	1	4
3	The speed of radio waves in free space is approximately _____ $\times 10^8$ m/s.	1	1	4
4	The sensor used to measure oxygen content in exhaust gases is called the _____ sensor.	1	1	1
5	The crankshaft position sensor provides information about engine _____.	1	1	1
6	The boost pressure sensor is mainly used in _____ engines.	1	1	1
7	The coolant temperature sensor monitors the temperature of engine _____.	1	1	1
8	MAF stands for Mass Air _____.	1	1	1
9	The throttle position sensor measures the angle of the _____ valve.	1	1	1
10	Rain sensors are commonly used for automatic _____ operation.	1	1	1

TEST

#	Question	M	BT	CO
1	A connected vehicle transmits real-time location, speed, and engine health information to a remote service center. Explain the concept of telematics, radio transmission, signal path, and information exchange involved in this system.	10	3	4
2	Discuss the construction, working principle, applications, and advantages of oxygen sensors and crankshaft/camshaft sensors used in modern automobiles.	10	2	1
3	A turbocharged vehicle is experiencing poor acceleration and increased fuel consumption. Analyze the role of the boost pressure sensor and Hot Film Air Mass Flow Sensor in diagnosing the problem.	10	4	1
4	Explain the working principle, construction, and automotive applications of coolant temperature sensors, throttle position sensors, and rain/light sensors.	10	2	1
5	A modern vehicle exhibits erratic engine performance and intermittent communication faults between control modules. Discuss how telematics systems and automotive sensors assist in monitoring, diagnosis, and communication within the vehicle.	10	5	4

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DEPARTMENT OF
HUMANITIES AND SOCIAL SCIENCES

Date : 19.06.2026	**CIE – III –Improvement Test**	**Max. Marks : 10 + 50**
Semester : VI	**UG- Institutional Elective**	**Duration : 2 hrs**
Course Title: Applied Psychology for Engineers		**Course Code : HS266TEW**

Sl. No	Questions	M	BT	CO
Part A				**10 Marks**
1.1	Define the trait approach to personality.	2	L4	CO4
1.2	Mention the psychological impact of stress.	2	L2	CO1
1.3	Mention the importance of learning in human behaviour.	2	L3	CO4
1.4	What is counselling? Mention the need for counselling.	2	L3	CO3
1.5	What is the role of an organizational psychologist in improving employee performance?	2	L2	CO2
Part – B				**50 Marks**
2	A young professional is highly motivated to develop personal skills, achieve goals, and reach his full potential. He believes in self-growth, making independent choices, and finding meaning in life despite challenges. Using the humanistic approach to personality, explain how personality development occurs in this individual. Discuss the key concepts of the humanistic approach.	10	L1	CO1
3	A psychologist is assessing a client who finds it difficult to express personal feelings, emotions, and inner conflicts directly. The psychologist uses ambiguous images and incomplete scenarios to understand the client's underlying thoughts and personality traits. Based on the given case, explain how projective techniques help the psychologist in understanding the client's personality and discuss the different types of projective techniques used in personality assessment.	10	L2	CO2
4	A student feels anxious whenever entering the examination hall because of previous experiences of stress and fear during exams. Over time, the examination hall itself becomes associated with anxiety, even before the exam begins. Using the concept of classical conditioning, explain how the student developed anxiety towards the examination hall. Identify the components of classical conditioning involved in this situation.	10	L3	CO2
5	An organization observes that some employees are experiencing stress, low motivation, and difficulty in adapting to workplace challenges. To support employees and improve their performance, the organization introduces counselling services. Discuss the different approaches of counselling used to support employees.	10	L4	CO3

6	The company is recruiting new employees and wants to select candidates with suitable skills and personality traits. After the selection process, the company plans to provide training to improve employee performance and workplace skills. Explain the steps involved in the employee selection and training process.	10	L2	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	2	4	2	2	–	2	2	2	4	–	–
	Test	Max Marks	10	20	10	10	–	10	20	20	–	–	–

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**19th June 2026**	**Maximum Marks**	**10 + 50**
Course Code	**IM266TEQ**	**Duration**	**120 Mins**
Sem	**VI Semester**	**Improvement CIE**	
ELEMENTS OF FINANCIAL MANAGEMNT			

Note:

1. Answer all the Questions.
2. Avoid verbose answers and base them on your general knowledge. You should provide evidence of you having acquired the Subject Knowledge related to the course.

PART – A

Sl. No.	Questions	M	BT	CO
1	State the decision rule of NPV	01	L_2	CO3
2	Which technique of capital budgeting ignores cash flows after payback?	01	L_2	CO3
3	Which source of long-term financing increases financial risk?	01	L_2	CO4
4	Which regulator in the Indian Financial System approves IPOs?	01	L_1	CO1
5.	Which market of the two broad categories related to buying and selling of securities deals with resale of securities?	01	L_1	CO1
5.	What is an IPO?	01	L_1	CO1
6.	What are Retained Earnings?	01	L_1	CO3
7.	What is Permanent Working Capital?	01	L_2	CO3
8.	Define Conservative Policy in the context of working capital policy?	01	L_2	CO3
9.	Which technique of capital budgeting may give multiple rates?	01	L_2	CO3
10.	Which source of long-term financing has a tax shield?	01	L_2	CO4

PART – B

Sl. No.	Questions	M	BT	CO
1.a	Based on the Information Provided below Project A: Cost ₹1,00,000; inflows ₹40,000 per year for 4 years. Project B: Cost ₹1,00,000; inflows ₹20,000 for 2 years, then ₹70,000 for 2 years. Discount rate = 10%. Which of the projects should be selected based on the NPV decision rule? Rework the problem based on the IRR Rule? Discuss NPV vs IRR Conflict.	10	L_4	CO3
2a.	Mention the major sources of long-term finance with their advantages and limitations.	06	L_2	CO4
b.	Explain the advantages and limitations of debentures as a source of finance.	04	L_2	CO4
3a.	Discuss the core ideas and concepts related to financing mix for working capital.	06	L_3	CO3
b.	What is book building in IPOs?	04	L_2	CO1
4a.	Discuss the determinants of working capital in manufacturing Vis-à-vis Service firms.	05	L_2	CO3

b.	Given Cost = ₹3,00,000. Cash inflows: ₹1,00,000 per year for 4 years. Discount rates: 10% and 15%. PV factors (10%, 4 years) = 3.17; (15%, 4 years) = 2.855. Compute the NPV using Discount Rates of 10% and 15%.	05	L_3	CO3
5a.	Explain the trade-off between Liquidity vs. profitability in financial management and working capital policy.	05	L_3	CO3
b.	Explain the role of the secondary market in price discovery.	05	L_3	CO1

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Department of Civil Engineering

Date	**19 June 2026**	**Maximum Marks**	**50+10**
Course Code	**HS266TEY**	**Duration**	**90+20 Min**
Sem	**VI**	**IMPROVEMENT TEST**	
Universal Human Values – III			

Sl. No.	Quiz	M	BT	CO
1	Nature is made up of _____ and _____.	02	1	2
2	Co-existence is ever-present, _____ & _____.	02	1	2
3	Which are the four orders in which coexistence is expressed?	02	1	1
4	Unit is _____ being in space. So, it has definite _____ in space.	02	2	2
5	What is the human goal at the level of self?	02	2	1

Sl. No.	Test	M	BT	CO
1	Explain the distinct roles of self-exploration, self-awareness, Self-evaluation in the process of inner evaluation.	10	2	2
2	Examine the statement " when realization, understanding and contemplation in the self, human behavior and work naturally become harmonious"	10	2	3
3	How does achieving harmony inside the self naturally lead to the smooth functioning and nurturing of the physical body.	10	3	2
4	Differentiate between the self and body based on their needs and activities.	10	3	3
5	What is the role of human being in this ever-expressing co-existence?	10	2	4

Course outcomes:

CO 1	Understand the basic human aspiration with program of its fulfillment and meaning of resolution in the complete expanse of human living
CO 2	Understand human being in depth and see how self is central to human being
CO 3	Understand existence in depth and see how coexistence is central to existence
CO 4	Understand human conduct and the holistic way of living leading to human tradition

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	04	26	20	10	06	34	20	–	–	–

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Academic year 2024-2025 (EVEN Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**25th June 2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM266TEQ**	**Duration**	**120 Mins**
Sem	**VII Semester**	**CIE – III**	
ELEMENTS OF FINANCIAL MANAGEMENT			

Note:
1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.

PART – A

S No	Questions	M	BT	CO
1	what are the key differences between debt and equity?	02	L_1	CO1
2	How should a business person use debentures as an instrument as opposed to venture capital?	02	L_2	CO1
3	List any three sources of internal accruals	02	L_1	CO1
4	Mention two disadvantages of raising funds through an IPO.	02	L_1	CO1
5	Who issues Government securities market and for what purpose?	02	L_1	CO1

PART – B

S No	Questions	M	BT	CO
1.	What is preference capital? Explain its key features in brief.	10	L2	CO1
2.	Write short notes on i. Initial Public offer ii. Debentures iii. Term Loans iv. Investment Banking	10	L3	CO1
3	Differentiate between the primary market and secondary market with examples.	10	L2	CO2
4.	What are the key features of venture capital. How do they differ from traditional financing methods.	10	L2	CO3
5.	Differentiate between Operating Cycle and Cash Cycle. Explain their significance in working capital management with suitable examples.	10	L3	CO1

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DEPARTMENT OF
HUMANITIES AND SOCIAL SCIENCES

Date : 25.06.2025	**CIE– Improvement Test**	**Max. Marks : 10 + 50**
Semester : VI	**UG- Institutional Elective**	**Duration : 30 + 90 Min**
Course Title: Applied Psychology for Engineers		**Course Code : HS266TEW**

Note:
Answer all the Questions

Sl. No	Questions	M	BT	CO
	Part – A			
1	Every time Riya touches a hot utensil, she quickly pulls her hand back. Over time, she learns to avoid touching them altogether. Which type of learning is illustrated in Riya's case?	2	L1	CO1
2	Seema was unable to switch on her new blender as it had no visible button. After examining it for a while, she suddenly realized that it turned on when the lid was locked in place. Identify the type of learning involved and mention two strategies used.	2	L2	CO2
3	Mention the role of psychologists in the IT environment.	2	L2	CO3
4	What are the information technology tools that are used in distance learning?	2	L2	CO4
5	Nina has been feeling constantly stressed due to academic pressure and finds it hard to manage her emotions. She approaches the college counsellor for help. What type of psychological counselling does Nina require?	2	L1	CO3
	Part – B			
1	Every time Meena took her dog out, she used a bell to signal feeding time. Soon, the dog began salivating just at the sound of the bell, even when no food was present. Apply the concept of classical conditioning to explain the dog's behavior. Identify the UCS, UCR, CS, and CR, and describe any two principles of classical conditioning.	10	L2	CO1
2	An employee was scolded by his manager each time he arrived late to work. Eventually, he started coming on time to avoid being scolded. Using the concept of Operant Conditioning, explain Rahul's change in behavior.	10	L2	CO2
3	Ravi recently joined a new office and felt unsure about how to interact during team meetings. Over the next few weeks, he carefully observed his senior colleagues—how they presented ideas, maintained body language, and responded to feedback. Gradually, Ravi began to adopt similar behaviors and felt more confident participating in discussions. Based on the above case, explain how observational learning operates in social and professional settings.	10	L3	CO2

4	An IT company is expanding its workforce and wants to hire psychological professionals to support employee well-being and improve workplace mental health. The HR manager is responsible for selecting and training the right candidates for this role. Explain the steps involved in the selection and training of psychological professionals to work in an organization, using the case above as context.	10	L2	CO2
5	What is the basic need for Counselling? Identify and explain the type of counseling the case below is referring to. An Engineering student has fallen ill twice before his Mathematics tests. He slowly developed anxiety that his health would fail, and so strongly believes and has a fear that something bad would happen before his exams, and refuses to attend the exams.	10	L4	CO2

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Department of Electronics and Communication Engineering

Date: 25.6.2025	**Improvement Test**	**Max. Marks: 50**
Semester: VI	**UG**	**Duration: $1\frac{1}{2}$ Hrs $+\frac{1}{2}$ Hrs**
Course Title: Human Machine Interface		**Course Code: EC266TEH**

S No	Questions	M	BT	CO
1.a.	Summarize the different types of HMI Data Analytics.	4	1	1
1.b.	What is HMI Profiling? Discuss the different types profiling parameters with an example.	6	2	1
2.a.	Discuss different types of haptic feedback and their applications.	4	2	2
2.b.	Illustrate the Bosch Graphic Test System with a neat block diagram. Also discuss the features of the same.	6	2	1
3.a.	List the different types of HMI Debugging. Discuss the different steps followed for Debugging methods.	5	2	2
3.b.	Discuss the different types of challenges in HMI testing.	5	2	2
4.a.	Write HTML code to create a **web-based dashboard** that displays: • Machine ID • Temperature (in °C) • Pressure (in Bar) • Motor Status (Running/Stopped)	4	2	1
4.b.	With a neat block diagram, explain gesture based HMI for Industrial Applications.	5	3	2
5.a.	Explain the system architecture used in haptic technology with a neat block diagram.	6	3	3
5.b.	Discuss the integration of PLC, motion control, and HMI within the TwinCAT environment.	5	1	1

S No	Improvement - Quiz	M	BT	CO
1.	TwinCAT supports _____ standard programming languages.	1	1	1
2.	_____ is the correct CSS syntax to change the background color of an element to blue.	1	2	1
3.	_____ is the correct HTML for creating a hyperlink.	1	2	2
4.	Which testing capabilities are included in the Bosch Graphics Test System?	1	1	1
5.	_____ diagnostic tool is typically used to verify IP-level connectivity between an HMI and a controller.	1	1	1
6.	_____ Unicode encoding is widely used to support multiple languages in HMI software.	1	2	1
7.	Haptics feedback can be realized in the form of Force, _____, and _____.	1	2	3
8.	Unity is a cross-platform game engine developed by _____.	1	1	1
9.	_____ metric is often measured during HMI profiling to assess …	1	1	1

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Department of Chemistry

Date: 25-06-2025	**Quiz**	**Max. Marks: 10**
Semester: 6	**UG**	**Duration: 30 min**
Course Title: ADVANCED ENERGY STORAGE FOR E-MOBILITY		**Course Code: CM266TEG**

SI NO	Questions	M	BT	CO
1	What is regenerative braking?	1	1	1
2	Give an example for reserve battery.	1	2	1
3	Identify the effect of molar mass on the capacity of the battery.	1	2	2
4	Name a lithium based primary battery.	1	3	1
5	Which type of electrolyte is used in methanol fuel cell?	1	1	1
6	Pseudo capacitors are superior than that of EDLC type super capacitor. Justify	1	5	4
7	Solar cell hybrid vehicles are not a good alternative for gasoline vehicles. Justify.	1	5	4
8	Give general representation of hydrogen oxygen fuel cells.	1	3	2
9	In fuel cell, design of porous electrode along with catalyst is challenging. Justify.	1	5	4
10	List any two major advantage of supercapacitor hybrid vehicles.	1	2	2

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Department of Chemistry

Date: 25-06-2025	**Improvement Test**	**Max. Marks: 50**
Semester: 6	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
Course Title: ADVANCED ENERGY STORAGE FOR E-MOBILITY		**Course Code: CM266TEG**

S No	Questions	M	BT	CO
1	Discuss the salient feature of the following along with neat labelled diagram/flowchart. i)Plug in hybrid vehicles ii) Fuel cell hybrid vehicles	10	3	3
2	Analyse any five battery characteristics with respect to advanced battery used in e mobility.	10	2	1
3	Outline the construction and working mechanism of Pseudo supercapacitor and EDLC. Along with neat labelled diagram how they help in sustainable transport in super capacitor vehicles?	10	4	2
4	By applying the knowledge of semiconductors, Illustrate construction and working of amorphous silicon solar cell along with neat labelled diagram. How these solar cells are used to construct of solar cell hybrid vehicles? Explain the working of it with flow chart.	10	3	2
5	Methanol based fuel cell vehicles are good alternative to gasoline vehicles. Justify. Explain the construction and working of methanol oxygen fuel cell along with their advantages and limitations.	10	2	2

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Department of Civil Engineering

Date: 25-06-2025	Improvement CIE	Max. Marks: 60
Semester: VI	UG	Duration: 2 Hrs
Course Title: Intelligent Transportation System		Course Code: CV266TEE

Quiz

S No	Questions	M	BT	CO
1.	Mention any two advanced traffic management measures used in urban areas.	2	2	3
2.	List any two objectives of a Traffic Management System	2	1	4
3.	List two key technologies used in Advanced Traveller Information Systems (ATIS).	2	1	3
4.	Summarize the development stages of a Traffic Management System in brief.	2	2	3
5.	State any two benefits of implementing Advance Vehicle Control Systems in modern vehicles.	2	2	4

S No	Questions	M	BT	CO
1.	Discuss the objectives and strategies involved in the development of an efficient Traffic Management System in smart cities.	10	3	3
2.	Design a functional layout of a Traffic Management Centre for a tier-2 city and explain the systems integrated within it.	10	5	3
3.	Design an outline for the development of a city-wide ITS-based Traffic Management Centre integrating ATMS, ATIS, and APTS.	10	5	4
4.	Evaluate the role of ITS in enhancing road safety and traffic efficiency in metropolitan cities.	10	4	3
5.	Analyze the technical and operational challenges faced in implementing AVCS in developing countries and propose feasible solutions.	10	3	4

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Department of Computer Science and Engineering

Date: June 2025	**Test - 3**	**Max. Marks: 60**
Semester: VI	**UG**	**Duration: 2 Hrs**
Course Title: ROBOTIC PROCESS AUTOMATION		**Course Code: CS266TED**

Sln	PART A Questions	M	BT	CO
1.	Name any two UiPath activities used for email automation.	02	2	3
2.	A _____ in Orchestrator is used to automatically start jobs at specific times or events.	01	2	3
3.	In email automation, attachments can be saved using the _____ property of the email message object.	01	1	3
4.	What is the key difference between the 'Breakpoint' and 'Log Message' activities, and how do they complement each other during debugging?	02	1	4
5.	Mention two core technologies commonly used in hyperautomation.	02	2	4
6.	A _____ is a debugging tool used to pause execution at a specific point to examine variables and flow.	01	2	3
7.	Robots are connected to Orchestrator using a Machine Key and _____.	01	1	3
	PART B Questions			
1. a	Explain the role of email protocols in email automation. How are these protocols used in UiPath? Discuss various associated activities in UiPath.	10	2	3
2. a	Differentiate between system exceptions and business exceptions in UiPath. Provide examples	06	2	3
b	What are the key debugging tools available in UiPath Studio? How do they assist in error resolution?	04	3	3
3. a	Explain the structure and function of Try-Catch activity in UiPath. How does it help in managing exceptions?	04	3	3
b	What is UiPath Orchestrator? Describe its main functionalities in the context of deploying and managing bots	06	2	3
4. a	Discuss the role of queues, assets, and logs in UiPath Orchestrator. How do they support bot management and execution?	06	3	4
b	Define hyperautomation and discuss its key components.	04	2	4
5. a	List and explain the key phases of hyperautomation. Illustrate any two real-world use cases of hyperautomation	05	3	4
b	Explain the concept of Hyperautomation-as-a-Service (HaaS). What are its benefits for small and medium-sized enterprises (SMEs)?	05	2	4

Course Outcomes

CO1	Understand RPA principles, its features and applications
CO2	Demonstrate proficiency in handling variables and decision making inside a workflow and data manipulation techniques.
CO3	Gain insights into recording, email automation and exception handling and orchestrator
CO4	Analyze the trends in automation and choose business strategy to design a real world automation workflow.

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

BT & CO Marks	L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4	CO5
	4	37	19	–	–	–	–	–	36	24	–

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Department of Mathematics

Date: 25-06-2025	**CIE 3**	**Max. Marks: 10+50**
Semester: VI	**UG**	**Duration: 2 Hrs (2PM - 4PM)**
Course Title: Mathematics of Quantum Computing		**Course Code: MA266TEV**

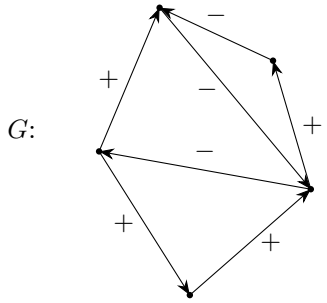
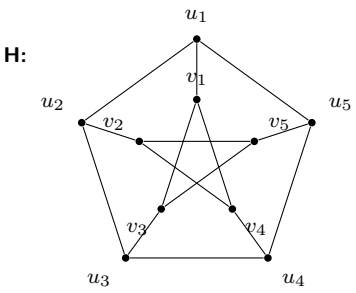
S No	Quiz	M	BT	CO
1	If the estimated phase in Quantum Phase Estimation algorithm is $\theta = 2\pi\left(\frac{3}{8}\right)$, what will be the output of the top register in binary?	2	2	2
2	In Shor's algorithm, what is the classical problem reduced to after finding the period r of a^x mod N?	2	1	1
3	What is the effect of the Grover diffusion operator on the state vector?	2	2	3
4	Describe in one sentence how Quantum Phase Estimation is used in both Shor's and HHL algorithms.	2	3	3
5	What quantum algorithm is used as a subroutine in Shor's algorithm to find the period of a function?	1	2	2
6	In the controlled rotation step of the HHL algorithm, what is encoded into the amplitude of the ancilla qubit?	1	1	1
S No	**Test**	**M**	**BT**	**CO**
1	State the problem that the Quantum Phase Estimation algorithm is designed to solve. Describe the complete quantum circuit used for QPE, including the role of each register and gate explaining step by step how the algorithm estimates the phase.	10	2	2
2	What is the significance of Shor's algorithm in cryptography? Explain how the period r is found out in this algorithm, thereby leading to determining the factorization of the given number N.	10	3	3
3	Outline the working of Grover's search algorithm explaining the reflections involved. Also analyse the number of iterations involved.	10	3	4
4	What are the key assumptions required for the efficiency of HHL algorithm? How does the algorithm solve the quantum version of the problem $A\vec{x} = \vec{b}$? Outline the high-level steps.	10	4	3
5	For a 2 qubit search space with one marked state $\|11\rangle$, compute the state after one iteration of the Grover operator starting from the uniform superposition state, and verify the probability of measuring the marked state.	10	4	4

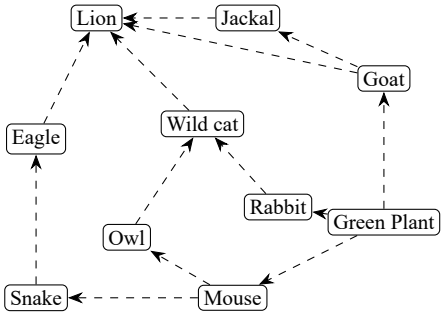
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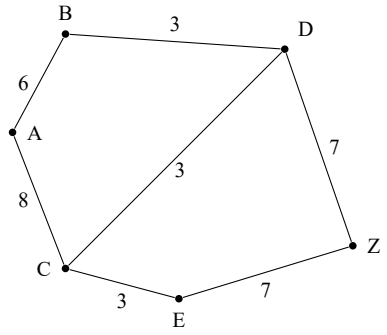
Department of Mathematics

Date: 25/06/2025	**Improvement CIE**	**Max. Marks: 100**
Semester: VI	**UG**	**Duration: 2 Hrs. (2pm-4pm)**
Course Title: Mathematical Modelling		**Course Code: MA266TEU**

Quiz

SL. No	Questions	M	BTL	CO
1	Is the given graph G balanced? Find the degree of balance of G: 	2	2	2
2	Draw the complete graph $K_{3,4}$.	2	1	1
3	Model a real life scenario using a digraph.	2	1	1
4	Define bipartite graph with an example.	2	3	2
5	The chromatic number of the following graph H. 	2	3	3

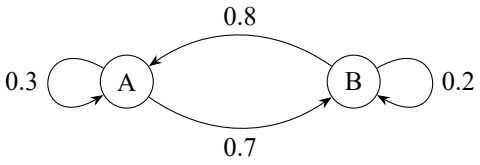
SL. No	Test	M	BTL	CO
1.	Define the Harary measure of status. Find the Harary measure of each individual in the organizational graph shown in the figure below. 	10	1	1

2.	The Mathematics department of a certain college plans to schedule the classes Graph Theory (GT), Statistics (S), Linear Algebra (LA), Advanced Calculus(AC), Geometry (G) and Modern Algebra (MA) this summer. Ten students given below have indicated the course they plan to take. With this information, use graph theory to determine the minimum number of time periods needed to offer these courses so that every two classes having student in common are taught at different time periods during the day. Of course, two classes having no students in common can be taught during the same period. Anden: LA,S, Brynn: MA,LA,G, Chase: MA, G,LA, Denise: G, LA, AC Everett: AC, LA, S, Francois: G,AC, Greg: GT, MA, LA Harper: LA, GT,S Irene: AC, S, LA Jennie: GT, S	10	2	3
3.	At the end of the school day, all the students plan on driving from school (Vertex A) to the concert (Vertex Z). The edges in the graph below represent one-way roads, and the weights of each edge represents the number of vehicles (in hundreds) that particular road can handle in one hour. What is the greatest number of vehicles that can get from school. 	10	3	3

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Department of Mathematics

Date: 30/06/2025	**Re-CIE**	**Max. Marks: 100**
Semester: VI	**UG**	**Duration: 2 Hrs. (2pm-4pm)**
Course Title: Mathematical Modelling		**Course Code: MA266TEU**

S No	Quiz: Questions	M	BT	CO
1	The angle of projectile to attain the maximum horizontal distance should be _____ the maximum height should be _____.	2	L1	2
2	A hot object at 120°C is placed in a room with temperature 30°C. If the cooling constant is $k = 0$, what is the temperature after 5 minutes?	2	L1	1
3	In the long run, the proportion of time that the given Markov chain will be in state A is _____ and state B is _____. 	2	L2	3
4	Define probability vector. Check whether the vector $\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0\right)$ is a probability vector or not.	2	L2	2
5	The growth of an organism follows the law $\frac{dx}{dt} = \frac{1}{13}x$. If initially there were 100 of them, then after 13 years, their approximate number will be _____ and their doubling period is approximately _____ years.	2	L1	1

Test

SL. No	Test: Questions	M	BT	CO
1.	A drug is induced in a patient's bloodstream at regular intervals of duration T each. Simultaneously the drug disappears at a rate proportional to the concentration $c(t)$ of the drug present at any time t. Determine the differential equation governing $c(t)$, solve it and interpret the results.	10	L3	2
2.	Write the systematic 12-step procedure for tackling a problem through the use of mathematical modelling.	10	L1	1
3a.	Describe the concept of a simple compartment model with an example.	05	L2	2
3b.	The territory of a sales man consists 3 cities. A, B, C. he never sells in the same city on successive days. If he sells in city A, then the next day he sells in B. However, if he sells either in B or C, then the next day he is twice likely to sell in city A as in the other city. How often does he sell in each of the cities in steady state?	05	L2	3
4a.	Using the limiting behaviour of the homogeneous Markov chain, find the steady state probability of the chain given by the TPM $P = \begin{bmatrix} 0.1 & 0.6 & 0.3 \\ 0.5 & 0.1 & 0.4 \\ 0.1 & 0.2 & 0.7 \end{bmatrix}$	05	L2	3
4b.	Explain linear and decay model with an example.	05	L1	1

5.	A gambler has ₹2. He bets ₹1 at a time and wins ₹1 with probability $\frac{1}{2}$. He stops playing if he ends up with no money or wins ₹4. a. What is the TPM of the related Markov chain? b. What is the probability that he has lost his money at the end of 5 plays? c. What is the probability that the game lasts more than 7 plays?	10	L3	4

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Academic year 2023-2024 (EVEN Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**28th August 2024**	**Maximum Marks**	**10 + 50**
Course Code	**21IE6F8**	**Duration**	**20 + 90 min**
Sem	**VI**	**Improvement CIE**	
Elements of Financial Management (Institutional Elective)			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Define the term Par Value.	01	–	–
2	What is Pre-emptive Right?	01	–	–
3	Mention any two phases of the IPO process.	01	–	–
4	What is Private Placement?	01	–	–
5	In which year the National Stock Exchange was established?	01	–	–
6	What is a Merger?	01	–	–
7	What is the role of World Bank in the International Financial System?	01	–	–
8	What is letter of Credit?	01	–	–
9	Define Hedging.	01	–	–
10	What type of forward contract is a futures contract?	01	–	–
	Part – B			
1	Enumerate the principle tasks of Securities Exchange Board of India (SEBI)? Describe briefly the key initiatives taken by SEBI.	10	–	–
2a	List the factors that have an important bearing on working capital needs?	05	–	–
b	Distinguish between operating cycle and cash cycle?	05	–	–
3a	Distinguish between Initial Public Offer (IPO) and Follow on Public Offer (FPO).	05	–	–
b	"Accruals are a 'free' source of finance". Analyze.	05	–	–
4	Discuss the plausible reasons for Mergers. In what way do you think mergers can enable organizations to beat competitions?	10	–	–
5	Illustrate the salient features of foreign exchange market in India. How does it differ from the features of international foreign exchange market?	10	–	–